MULTIMODAL AI HACKATHON 2026 | TRACK A

CardioMap 3D

Cardiovascular Risk Visualization & Prediction

Project documentation | Team Visionaries (Dream - Innovate - Build)

Hackathon	Multimodal AI Hackathon 2026
Track	Track A: Cardiovascular Risk Visualization and Prediction
Team name	Team Visionaries (Dream · Innovate · Build)
Team members	**Yogesh Mondhe** (Team Leader), **Khushi Wanjari** (Team Member)
Live demo	https://cardiomap-3d.vercel.app
Backend API docs	https://cardiomap-api.onrender.com/docs
Source code	https://github.com/yogeshx21/cardiomap-3d

Clinical safety notice: CardioMap 3D is an educational / decision-support prototype. It is not a medical device and is not a substitute for formal diagnostic imaging or a clinician's judgement.

Abstract

Heart disease risk is usually reported as a few numbers in a table. CardioMap 3D takes a patient's clinical information and shows the predicted risk on an **interactive 3D heart**. Each coronary artery (LAD, LCX and RCA) is colored from green (low risk) to red (high risk), and the app explains **why** a score was given using SHAP. Four machine learning models predict overall coronary artery disease (CAD) and narrowing in each of the three main arteries, using 59 clinical inputs from the Z-Alizadeh Sani dataset. The whole system is open, runs on free hosting, and needs no paid API.

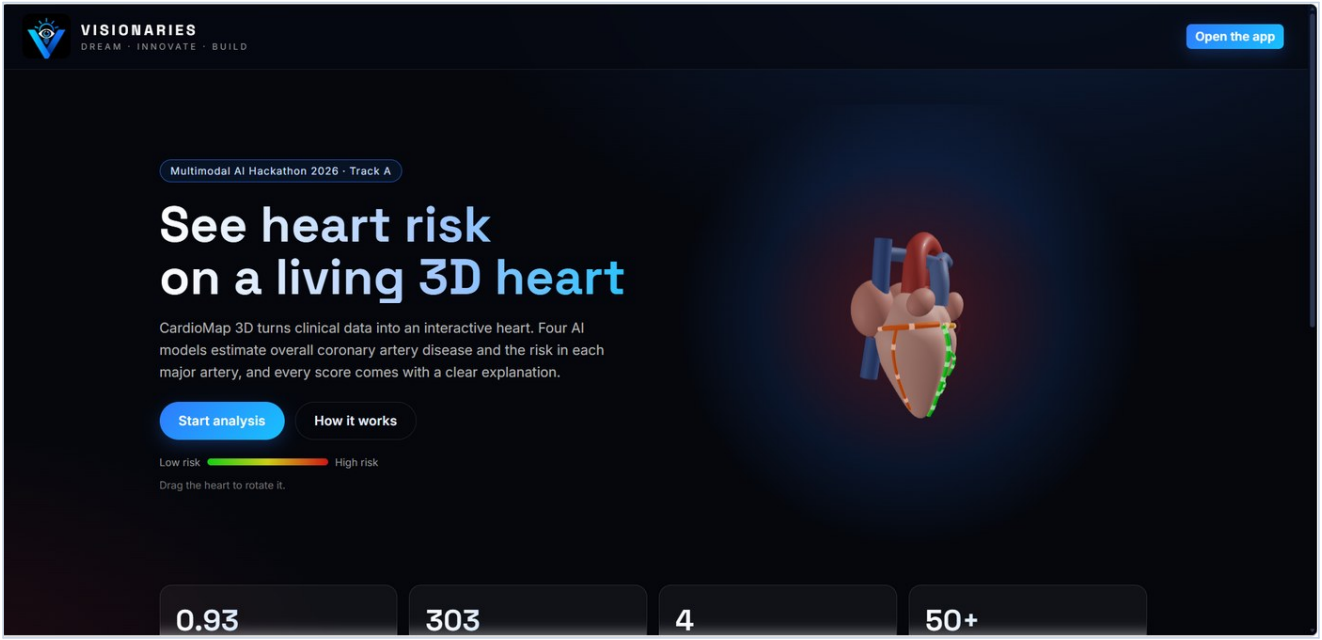


Figure 1: Landing page with a live 3D heart whose arteries change color on their own.

0.93	**303**	**4**	**59**
ROC-AUC for overall CAD (nested CV)	patients in the dataset	predictions: CAD, LAD, LCX, RCA	clinical input features

1. Problem statement

Heart disease risk is usually reported as a few numbers in a table, which is hard to interpret quickly. Doctors and students have to mentally connect "this patient's risk score" with "which coronary artery is probably affected".

2. Our solution

CardioMap 3D connects the numbers to the anatomy in three steps:

Step	What happens
1. Enter patient data	The user types values or loads one of three demo patients. Inputs cover vitals, symptoms, ECG, laboratory tests and echocardiography.
2. AI predicts four risks	Separate models estimate overall CAD and the narrowing risk in the LAD, LCX and RCA arteries.
3. See it on the heart	Each coronary artery changes color live, and the dashboard explains which factors pushed the score (SHAP).

Multimodal inputs and outputs

Input: demographics and risk factors, symptoms, ECG findings, laboratory tests and echocardiography measurements, combined in one prediction (59 features). **Output:** numbers (risk scores), explanations (SHAP factors) and a 3D visualization, plus an illustrative ECG strip, a heartbeat sound and a read-aloud voice summary.

3. Key features

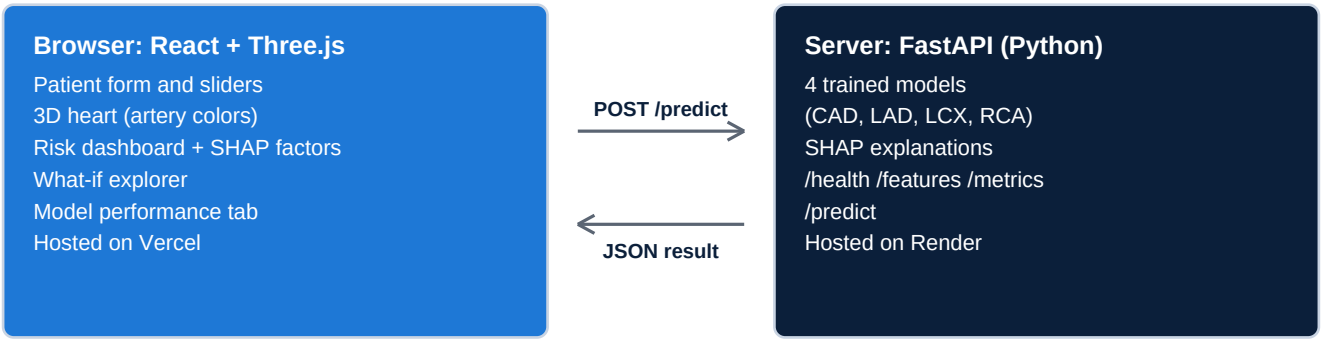
- 4 prediction targets: overall CAD, LAD, LCX and RCA
- Landing page with a live 3D heart
- Interactive 3D heart: rotate, zoom, X-ray mode, auto-rotate, reset view
- Click an artery (or its card): the camera flies to it and the others fade
- Artery colors update live (about 0.3 s after any input changes)
- Blood-flow pulses and a schematic narrowing that follow each artery's risk, plus a warning ring on high-risk arteries
- Heartbeat animation and optional sound that follow the pulse rate (PR)
- Illustrative ECG strip that reacts to T inversion, ST changes, Q wave and LVH
- Overall CAD risk gauge and a "Read aloud" voice summary
- What-if explorer: blood pressure, LDL, smoking, BMI, blood sugar
- Three demo patients and an automatic guided tour
- Explainable AI: top factors per patient and global SHAP importance; Model performance tab; leakage-safe training

4. Who can use it and why

User	Use
Medical students	Learn which risk factors drive coronary disease and how they relate to specific arteries
Clinicians (decision support)	Quick visual overview of a patient's risk profile
Patients / public education	Understand risk factors through a visual, interactive model
Researchers / developers	Reusable pipeline: training, explainability and a web API

5. System architecture

The web app (React + Three.js) sends the patient's inputs to our own FastAPI backend. The backend fills any missing features with the training median, runs the four models, computes SHAP explanations and returns a JSON result that the 3D heart and dashboard display.



GitHub (main branch) redeploys both Vercel and Render automatically

6. Tech stack

Languages: Python 3.14, JavaScript (React JSX), HTML, CSS.

Machine learning and backend	Version	Used for
scikit-learn	1.9.1	Logistic Regression, Random Forest, preprocessing, cross-validation
XGBoost	3.4.1	Gradient boosting model (candidate)
SHAP	0.53.0	Explainability
pandas / NumPy / SciPy	3.0.6 / 2.5.3 / 1.18.1	Data handling
joblib	1.6.0	Saving and loading trained models
FastAPI / Uvicorn / Pydantic	0.143.0 / 0.54.0 / 2.14.0	REST API, server, request validation
openpyxl, matplotlib	-	Reading the Excel dataset, plots (training only)

Frontend and platforms	Used for
React + Vite	Web app and build tooling
Three.js, @react-three/fiber, @react-three/drei	3D graphics, React bindings, orbit controls and labels
Web Audio, Web Speech, Canvas 2D	Heartbeat sound, voice summary, ECG strip (built into the browser)
Vercel / Render (free web service) / GitHub	Frontend hosting / FastAPI hosting / source control and automatic deployments
Node.js 24, npm, Git	Frontend build and version control

The 3D heart is **procedurally generated in code** (no downloaded 3D model), so there are no licence issues and each artery maps directly to a model output.

7. REST API (our own, no paid third-party service)

Method	Endpoint	Purpose
GET	/health	Server status, loaded models, explanation method
GET	/features	Feature names, ranges and median values (used to build the form)
GET	/metrics	Model evaluation table
POST	/predict	Risk for CAD, LAD, LCX, RCA plus top SHAP factors

```
POST /predict
{ "features": { "Age": 63, "HTN": 1, "DM": 1, "BP": 150 } }
```

Features that are not sent are filled with the training median. The response contains, for each target, a probability, a risk level (low / medium / high) and the top contributing factors.

8. Dataset

Z-Alizadeh Sani dataset (extended version, UCI Machine Learning Repository): **303 patients** with **59 input features** covering demographics, symptoms, ECG, laboratory tests and echocardiography. The angiography outcomes (Cath, LAD, LCX, RCA) are the prediction targets and are never used as model inputs.

9. Machine learning pipeline

- **Cleaning:** no missing values; categorical columns encoded numerically.
- **Leakage control:** target columns (Cath, LAD, LCX, RCA) are removed from the inputs, and the code asserts this.
- **Models compared:** Logistic Regression, Random Forest and XGBoost for each of the 4 targets.
- **Evaluation:** nested 5-fold stratified cross-validation. Hyperparameters are tuned in an inner 3-fold loop and scores are measured on held-out outer folds, which gives honest estimates.
- **Final models:** the best model per target is tuned on the full data and saved (joblib) for the API.
- **Explainability:** SHAP values per prediction and global SHAP importance plots.

10. Results (nested cross-validation)

Target	Best model	ROC-AUC	Accuracy	Precision	Recall	F1
CAD (overall)	Logistic Regression	0.926	0.878	0.945	0.880	0.911
LAD	Random Forest	0.849	0.795	0.804	0.859	0.831
LCX	Random Forest	0.734	0.680	0.587	0.622	0.604
RCA	Random Forest	0.712	0.673	0.565	0.570	0.568

The numbers are reported as they came out of the nested cross-validation. They are lower than many published 99% results because leakage columns are removed and the evaluation is nested.

What the models rely on

For overall CAD the most influential features are **Typical Chest Pain, Age, Atypical chest pain, Region RWMA, hypertension, T-wave inversion, diabetes and dyspnea**, which is consistent with clinical knowledge. For LAD the top features are Typical Chest Pain, Region RWMA (wall motion) and EF-TTE (pumping strength). For LCX and RCA the importance values are small and scattered across lab values, which matches their weaker AUC (0.73 and 0.71), so these two arteries should be read as indicative only.

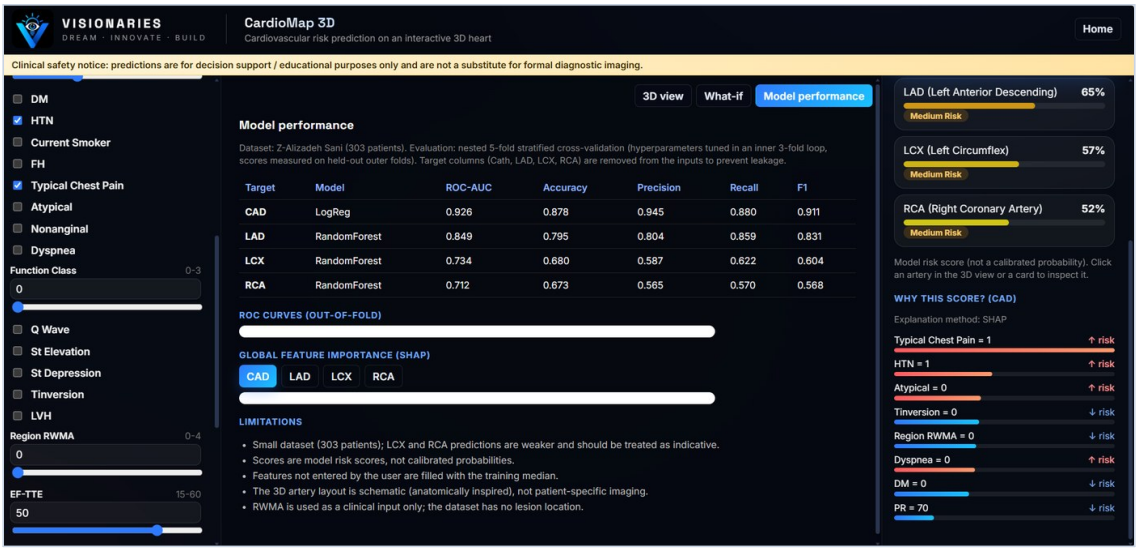


Figure 2: The "Model performance" tab inside the app: metrics table, ROC curves, global SHAP plots and limitations.

11. 3D visualization and app walkthrough

The heart (ventricles, atria, aorta, pulmonary trunk, vena cava) is built from mathematical surfaces in Three.js. LAD, LCX and RCA, with their main branches, are drawn as tubes on the heart surface. The probability maps to color: **green (low) to yellow (medium) to red (high)**. Clicking an artery (or its card) shows that artery's own explanation.

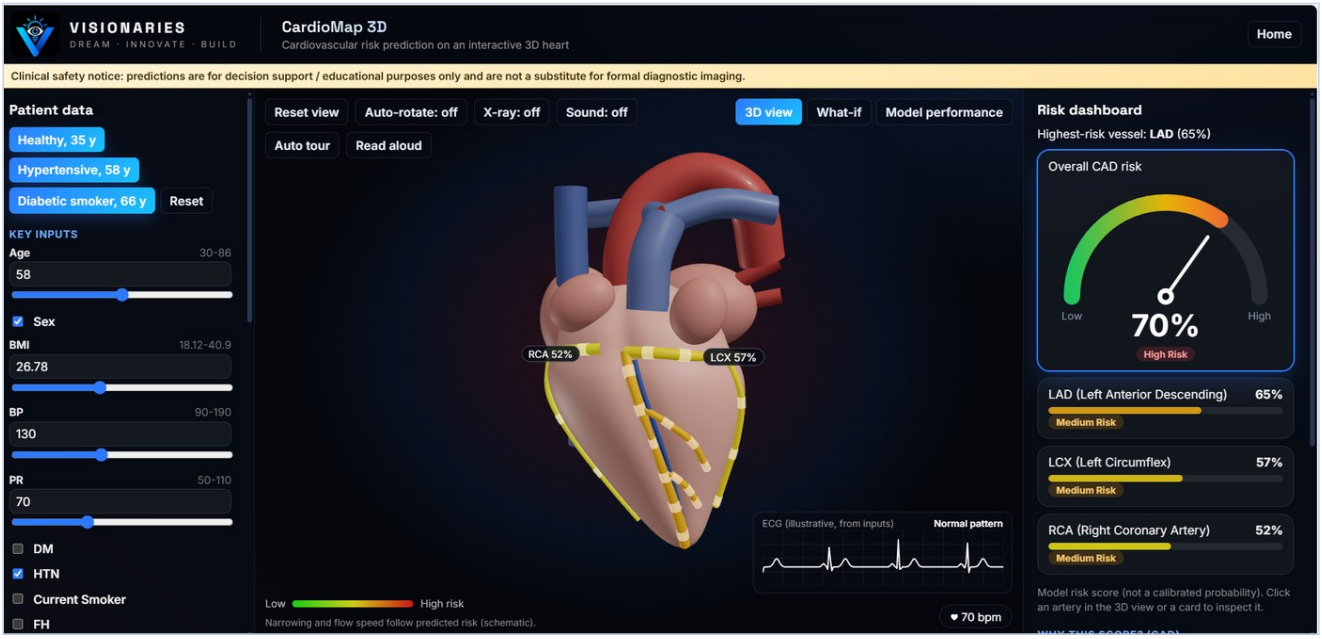


Figure 3: Main screen. Patient inputs on the left, interactive 3D heart with artery labels and illustrative ECG in the centre, risk dashboard (overall CAD gauge plus LAD / LCX / RCA cards) on the right.

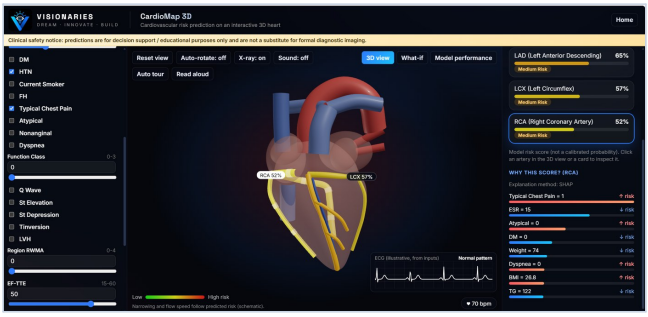


Figure 4: X-ray mode with a selected artery and its SHAP factors ("Why this score?").

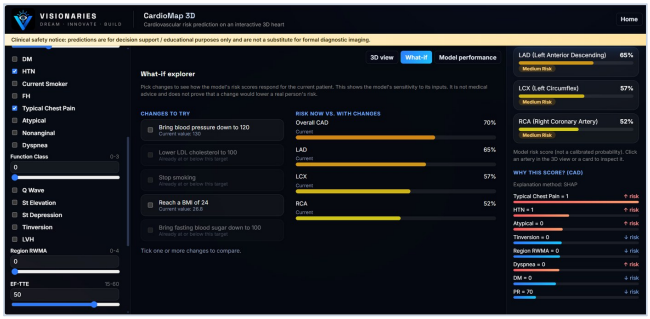


Figure 5: What-if explorer comparing current risk with changed inputs.

How to read the 3D view

- **Color and gauge:** each artery's color and the overall CAD gauge follow the model's risk score for that target.
- **Flow and narrowing:** moving flow pulses and a narrowed section whose depth follows the predicted risk. The location of the narrowing is illustrative only, because the dataset has a risk label per vessel but no lesion position.
- **Heartbeat and ECG:** the heartbeat animation follows the entered pulse rate (PR); the ECG strip is generated from the entered ECG findings and is not a real recording.
- **Demo and tour:** three demo patients (Healthy 35 y, Hypertensive 58 y, Diabetic smoker 66 y) and an Auto tour for a guided walkthrough; "Read aloud" speaks a summary of the result.

12. Limitations and safety

- The dataset is small (303 patients), so LCX and RCA predictions are weaker and should be treated as indicative only.
- Scores are model risk scores, **not calibrated probabilities**.
- Inputs the user does not provide are filled with the training median.
- The 3D artery layout is schematic (anatomically inspired), not patient-specific imaging.
- The dataset has no lesion location, so regional wall-motion abnormality (RWMA) is used only as an input feature.
- The ECG strip is generated from the entered findings; it is not a real recording.
- What-if results show how the model responds to changed inputs. They are not medical advice and do not prove that a change would lower a real person's risk.
- Not validated on external hospitals; not for real clinical decisions.

13. Setup and run (summary)

Prerequisites: Git, Python 3.14 (tested), Node.js 24 (tested; Node 20+ should work) and a recent Chrome or Edge. No API keys or paid services are needed. The full instructions are in the README of the GitHub repository.

```
git clone https://github.com/yogeshx21/cardiomap-3d.git
cd cardiomap-3d
python -m venv venv && venv\Scripts\activate
pip install -r backend/requirements.txt
cd frontend && npm install && cd ..

# Terminal 1 (repository root, venv active)
uvicorn main:app --app-dir backend --reload
# Terminal 2
cd frontend && npm run dev     # open http://localhost:5173
```

Check: http://127.0.0.1:8000/health returns "status":"ok" with the four models loaded. In the app, press "Diabetic smoker, 66 y" to see the arteries turn yellow or red.

14. Deployment

- **Backend:** Render web service. Build: `pip install -r backend/requirements.txt`; start: `uvicorn main:app --app-dir backend --host 0.0.0.0 --port $PORT`. The free tier sleeps when idle, so the first load can take about a minute.
- **Frontend:** Vercel with root directory `frontend` and the environment variable `VITE_API_URL` pointing to the Render URL. Optional `LITE_EXPLAIN=1` uses lightweight explanations on low-memory hosts.
- Pushing to the `main` branch on GitHub redeploys both automatically.

15. Possible next steps

- Train on a larger, multi-hospital dataset and validate externally, especially for LCX and RCA.
- Calibrate the probabilities so the scores can be read as real likelihoods.
- Support imaging inputs (for example angiography or echo images) as an additional modality.

16. Acknowledgements

Z-Alizadeh Sani dataset (UCI Machine Learning Repository). Open-source libraries: scikit-learn, XGBoost, SHAP, FastAPI, React and Three.js.

Team Visionaries (Dream · Innovate · Build): Yogesh Mondhe (Team Leader) and Khushi Wanjari (Team Member). Built for the Multimodal AI Hackathon 2026, Track A: Cardiovascular Risk Visualization and Prediction.